

Total No. of Questions : 8]

SEAT No. :

P7606

[Total No. of Pages : 2

[6180]-125

T.E. (Electronics and Computer Engineering)
MICROCONTROLLER AND APPLICATIONS
(2019 Pattern) (Semester-I) (310344)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data if necessary.*

- Q1)** a) Interface relay with 8051 microcontroller. Write an embedded C program to turn ON and OFF relay with delay. **[6]**
- b) Draw an interfacing diagram of stepper motor with 8051 microcontroller and write an embedded C code to rotate it in clockwise continuously. **[6]**
- c) Draw an interfacing diagram of DAC with 8051 microcontroller and write an embedded C code to triangular waveforms. **[8]**

OR

- Q2)** a) Draw an interfacing diagram of opto-isolator with 8051 microcontroller and write an embedded C code to flash lamp connected to it with delay of 10 msec. **[6]**
- b) Interface buzzer with 8051 microcontroller. Write an Embedded C program to turn ON and OFF buzzer. **[6]**
- c) Explain the interfacing of RS232 with 8051 with neat diagram. Also highlight the role of MAX232 **[8]**

- Q3)** a) State any 4 differences between MSP430x2x, MSP430x4x, MSP430x5x. **[8]**
- b) Explain with neat diagram MSP430x5xx microcontroller architecture. **[8]**

OR

- Q4)** a) Compare variants of the MSP430 family. **[8]**
- b) Explain in detail register set used in MSP430 microcontroller **[8]**

P.T.O.

- Q5) a)** Explain the following GPIO registers of MSP430 microcontroller. **[8]**
- i) PxIES
 - ii) PxREN
 - iii) PxOUT
 - iv) PxDIR
- b) Draw an interfacing diagram of 8 LEDs with MSP430 microcontroller and write an embedded C code to flash the LEDs with is delay. **[8]**

OR

- Q6) a)** Write a short note on 12C and UART protocol of MSP430 microcontroller. **[8]**
- b) Explain in detail PWM generation with various duty cycle in MSP430x5xx **[8]**

- Q7) a)** Design frequency counter using 8051 microcontroller and display the result on LCD. **[6]**
- b) Design a home automation system using MSP430 microcontroller. Also write an embedded C program for same. **[6]**
- c) Draw and explain block diagram of DAS using 8051 microcontroller. **[6]**

OR

- Q8) a)** Design environment monitoring system using MSP430 microcontroller. **[6]**
- b) Design soil monitoring system for agriculture using MSP430 microcontroller. **[6]**
- c) Design a water level monitoring system using 8051 microcontroller. Draw the flow chart for the same. **[6]**



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SEAT No. :

PB-3847

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[6262]-109

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- Q1)** a) Draw interfacing diagram and write a C program for the 8051 microcontroller to transfer the letter "A" serially at 4800 baud continuously. Use 8 bit data and 1 stop bit. [6]
- b) Draw interfacing diagram of opto - isolator with 8051 microcontroller. Also write an Embedded C program for same. [6]
- c) Explain the working principle of motion detector (PIR) sensor. Draw interfacing diagram of motion detector sensor and buzzer with 8051 microcontroller and write an Embedded C program to turn on buzzer when motion is detected. [8]

OR

- Q2)** a) Interface buzzer with 8051 microcontroller. Write an Embedded C program for the same. [6]
- b) Draw interfacing diagram of DAC to 8051 microcontroller and write an Embedded C program to generate sine wave form. [6]
- c) Draw interfacing diagram of stepper motor with 8051 microcontroller. Write an Embedded C program to rotate stepper motor continuously in clockwise direction. [8]
- Q3)** a) Explain in detail any four addressing modes of MSP430 microcontroller. [8]
- b) Explain in detail: [8]
- i) Low power modes in MSP430 microcontroller.
 - ii) Compare FRAM and Flash memory

P.T.O

OR

Q4) a) State the function of following registers of MSP430 microcontroller : **[8]**

- i) Status Register
 - ii) Program Counter
 - iii) Stack pointer
- b) List features and applications of variants of MSP430 family MSP430x2x, MSP430x4x, MSP430x5x. **[8]**

Q5) a) Write an Embedded C program for blinking of LED with and without switch using MSP430. Also draw its interfacing diagram. **[8]**

- b) Interface relay with MSP430 microcontroller. Write an Embedded C program for the same and draw its flowchart. **[8]**

OR

Q6) a) Draw interfacing diagram of IR sensor with MSP430 microcontroller. Also write the Embedded C program for the same along with its flowchart. **[8]**

- b) Draw and explain interrupt structure of MSP430. **[8]**

Q7) a) Explain with neat block diagram DAS using 8051 microcontroller. **[9]**

- b) Design environment Monitoring System using MSP430 microcontroller. Also draw the flowchart for same. **[9]**

OR

Q8) a) Design Home Automation System using MSP430 microcontroller. Also draw the flowchart for same. **[9]**

- b) Design frequency counter using 8051 microcontroller which will display the result of various frequency ranges on LCD. Also write the algorithm for same. **[9]**



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- b) Design soil monitoring system for agriculture using MSP430 microcontroller. **[6]**
- c) Design a water level monitoring system using 8051 microcontroller. Draw the flow chart for the same. **[6]**

